PLANE

PLANE: A plane is a surface such that if any two points are taken on it, the straight line joining them lies wholly on it.

General equation of a plane: prove that the general equation of the first degree in x, y, z i.e. $ax+by+cz+d=0$ represents a plane.

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be the co-ordinates of any two points lying on the locus whose eqn is

$$ax+by+cz+d=0 \quad \text{--- (1)}$$

$$\therefore ax_1+by_1+cz_1+d=0 \quad \text{--- (2)}$$

$$\text{and } ax_2+by_2+cz_2+d=0 \quad \text{--- (3)}$$

Multiplying (3) by K and adding with (2), we get

$$a(x_1+Kx_2)+b(y_1+Ky_2)+c(z_1+Kz_2)+d(1+K)=0$$

$$\Rightarrow \frac{a(x_1+Kx_2)}{1+K}+\frac{b(y_1+Ky_2)}{1+K}+\frac{c(z_1+Kz_2)}{1+K}+d=0 \quad \text{--- (4)}$$

The eqn (4) clearly shows that the point $\left(\frac{x_1+Kx_2}{1+K}, \frac{y_1+Ky_2}{1+K}, \frac{z_1+Kz_2}{1+K}\right)$ lies on (1) for all values of K.

Thus it has been shown that if the points (x_1, y_1, z_1) and (x_2, y_2, z_2) lies on (1), then this point dividing the join of them in any ratio K:1 also lies wholly on (1). In other words the line joining (x_1, y_1, z_1) and (x_2, y_2, z_2) lies wholly on (1). Hence by the definition of the plane we conclude that (1) represents a plane.

Thus, the general eqn of the first degree in x, y, z i.e $ax+by+cz+d=0$ represents a plane.

Plane determined by a point and a normal vector

A plane in 3-space can be determined uniquely by specifying a point in the plane and a vector perpendicular to the plane (Fig 1). A vector perpendicular to a plane is called a normal to the plane.

Suppose that we want to find an eqn of the plane passing through $P_0(x_0, y_0, z_0)$ and perpendicular to the vector $\vec{n} = \langle a, b, c \rangle$. Define the vector $\vec{r}_0$ and $\vec{r}$ as

$$\vec{r}_0 = \langle x_0, y_0, z_0 \rangle \text{ and } \vec{r} = \langle x, y, z \rangle$$

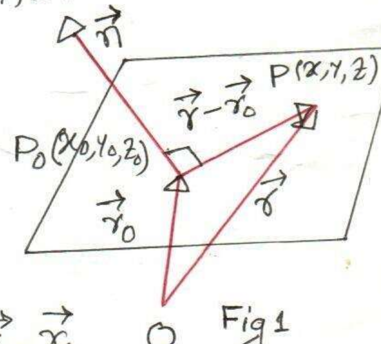


Fig 1

It should be evident from fig 1 that the plane consists precisely of those points $P(x, y, z)$ for which the vector $\vec{r} - \vec{r}_0$ is orthogonal to $\vec{n}$; or, expressed as an eqn

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$$

$$\Rightarrow \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

$$\Rightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

This is called the point-normal form of the eqn of a plane.

Ex: Find an eqn of the plane passing through the point $(3, -1, 7)$ and perpendicular to the vector $\vec{n} = \langle 4, 2, -5 \rangle$

Soln: The point-normal form of the eqn is

$$4(x - 3) + 2(y + 1) - 5(z - 7) = 0$$

$$\Rightarrow 4x + 2y - 5z + 25 = 0.$$

Every equation of the first degree in x, y, z represents a plane.

Cor 1. General eq^n of a plane that passes through a given point.

The general eq^n of a plane is $ax+by+cz+d=0$ --- (1)

Since it passes through the point (x_1, y_1, z_1), we have

$ax_1+by_1+cz_1+d = 0$ --- (2)

subtracting (2) from (1) we have

$a(x-x_1)+b(y-y_1)+c(z-z_1) = 0$

which is the required eq^n of the plane that passes through the point (x_1, y_1, z_1).

Ex: Find ~~the~~ eq^n of plane passes through $(1,2,3)$.

Cor 2. General eq^n of a plane passes through the origin

In case the plane passes through the origin $(0,0,0)$ the general eq^n of the plane $ax+by+cz+d=0$ will be satisfied by $(0,0,0)$ so that $d=0$. Hence the eq^n reduces to the form

$$ax+by+cz=0$$

Cor 3. The eq^n of a plane passing through the two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$(x-x_1)(y-y_2)-(x-x_2)(y-y_1) = A\{(x-x_1)(z-z_2)-(x-x_2)(z-z_1)\}$$

Ex: Find the eq^n of plane passes through $(1,2,3)$ and $(2,3,4)$

Plane passing through three given points.

Let (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) are three non-colinear points. We have to find an eq^n of plane that passes through these 3 points.

Let the eq^n of the plane be

$ax+by+cz+d=0$ --- (1)

Since the given points lie on the plane ①, we must have

$$ax_1 + by_1 + cz_1 + d = 0 \quad --- ②$$
$$ax_2 + by_2 + cz_2 + d = 0 \quad --- ③$$
$$ax_3 + by_3 + cz_3 + d = 0 \quad --- ④$$

Eliminating a, b, c, d from ①, ②, ③ and ④ we have

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

Ex: Find the eqn of plane passes through $(-1,2,3)$, $(2,-3,4)$ and $(-3,4,5)$.

Ans: $19x + 14y + 13z - 48 = 0$

☐ Condition for four points may be coplanar is

The four points (x_1,y_1,z_1), (x_2,y_2,z_2), (x_3,y_3,z_3) and (x_4,y_4,z_4) will be coplanar if any plane through any three points passes through the fourth point also.

Now the plane passes through the points (x_2,y_2,z_2), (x_3,y_3,z_3) and (x_4,y_4,z_4) is

$$\begin{vmatrix} x & y & z & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0 \quad --- ①$$

The given four points will be coplanar if (x_1,y_1,z_1) lies on the above plane ①, i.e.

$$\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

Ex: Find the eqn of the plane passes through the points

(a) $(2,1,3)$, $(-1,-2,4)$, $(4,2,1)$

(b) $(1,1,-1)$, $(-2,-2,2)$, $(-1,-1,2)$

(c) $(2,1,-3)$, $(3,-1,4)$, $(7,5,6)$

Solⁿ: (a) The eqn of the plane passes through $(2,1,3)$, $(-1,-2,4)$, $(4,2,1)$ is

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 1 & 3 & 1 \\ -1 & -2 & 4 & 1 \\ 4 & 2 & 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow x\begin{vmatrix} 1 & 3 & 1 \\ -2 & 4 & 1 \\ 2 & 1 & 1 \end{vmatrix} - y\begin{vmatrix} 2 & 3 & 1 \\ -1 & 4 & 1 \\ 4 & 1 & 1 \end{vmatrix} + z\begin{vmatrix} 2 & 1 & 1 \\ -1 & -2 & 1 \\ 4 & 2 & 1 \end{vmatrix}$$

$$-1\begin{vmatrix} 2 & 1 & 3 \\ -1 & -2 & 4 \\ 4 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow x\{(4-1) - 3(-2-2) + 1(-2-8)\} - y\{2(4-1) - 3(-1-4) + 1(-1-16)\}$$

$$+ z\{2(-2-2) - 1(-1-4) + 1(-2+8)\} - 1\{2(-2-8) - 1(-1-16) + 3(-2+8)\} = 0$$

$$\Rightarrow x(3+12-10) - y(6+15-17) + z(-8+5+6) - 1(-20+17+18) = 0$$

$$\Rightarrow 5x - 4y + 3z - 15 = 0$$

OR $(x-x_1)(y-y_2) - (x-x_2)(y-y_1) = A\{(x-x_1)(z-z_2) - (x-x_2)(z-z_1)\}$

Ex: Show that the ~~four~~ points are coplanar

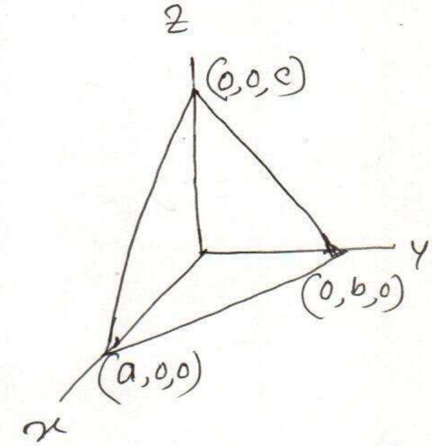
(a) $(0,4,3)$, $(-1,-5,-3)$, $(-2,-2,1)$, $(1,1,-1)$

(b) $(-6,3,2)$, $(3,-2,4)$, $(5,7,3)$, $(-13,17,-1)$

Some Standard forms of the eqn of a plane

(a) Intercept form: Find the eqn of a plane in terms of the intercepts, which its makes on the axes.

Let the intercepts made by the plane on the axes be a, b, c. Then we can regard it as a plane passing through the points $(a,0,0)$, $(0,b,0)$, $(0,0,c)$. Then the eqn is

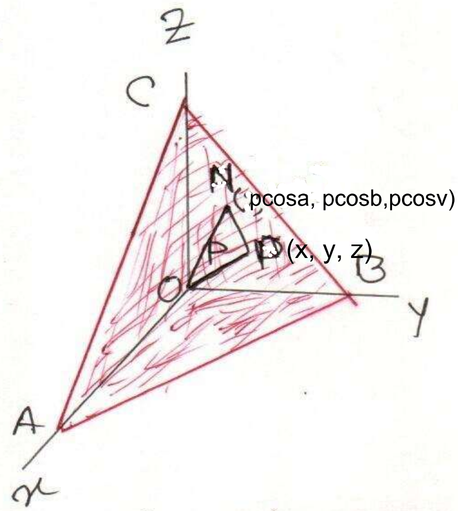


$$\begin{vmatrix} x & y & z & 1 \\ a & 0 & 0 & 1 \\ 0 & b & 0 & 1 \\ 0 & 0 & c & 1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

(b) Normal form: Find the eqn of a plane in terms of P, the length of the perpendicular from origin on the plane and the direction cosines l, m, n of this perpendicular.

Let ABC be the given plane and let ON be the normal from O to the plane ABC



Let the direction cosines of ON be $\cos\alpha$, $\cos\beta$, $\cos\gamma$ and Let $ON = p$

Then the co-ordinates of N are $(p\cos\alpha, p\cos\beta, p\cos\gamma)$.

Let $P(x,y,z)$ be any point on the plane ABC. Then NP is a line passing through N and lying on the plane ABC. Therefore ON and NP are at right angle

The direction cosines of NP are proportional to $x-p\cos\alpha$, $y-p\cos\beta$, $z-p\cos\gamma$.

Since ON and NP are at right angles, we must have

$\langle x-p\cos\alpha,\ y-p\cos\beta,\ z-p\cos\gamma\rangle\cdot\langle\cos\alpha,\cos\beta,\cos\gamma\rangle=0$

$\Rightarrow (x-p\cos\alpha)\cos\alpha+(y-p\cos\beta)\cos\beta+(z-p\cos\gamma)\cos\gamma=0$

$\Rightarrow x\cos\alpha+y\cos\beta+z\cos\gamma=p(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)=p$

$\Rightarrow x\cos\alpha+y\cos\beta+z\cos\gamma=p$ since $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$.

This eq^n, satisfied by the co-ordinates of every point on the plane, represents the plane.

If l, m, n be the direction cosines of the normal ON i.e. $l=\cos\alpha$, $m=\cos\beta$, $n=\cos\gamma$, then the eq^n of the plane ABC will be

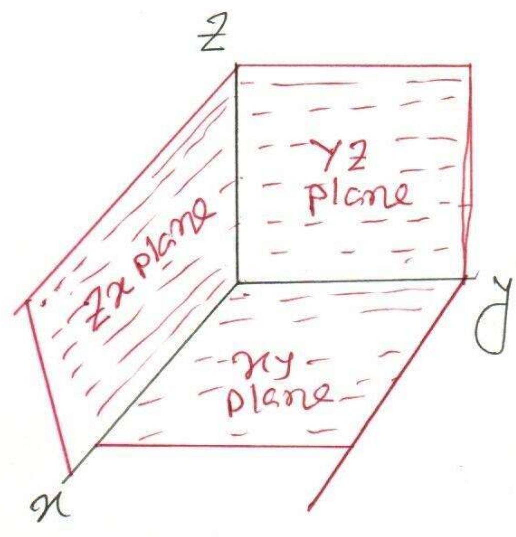
$lx+my+nz=p$.

Equations of the coordinate planes

(I) Equation of yz plane is $x=0$

(II) " " zx " " $y=0$

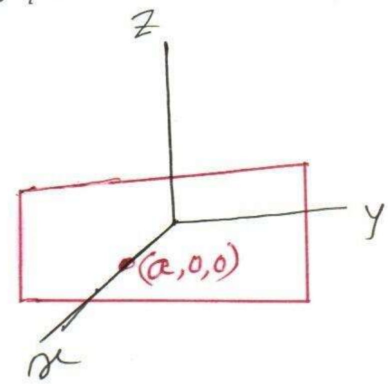
(III) " " xy " " $z=0$.

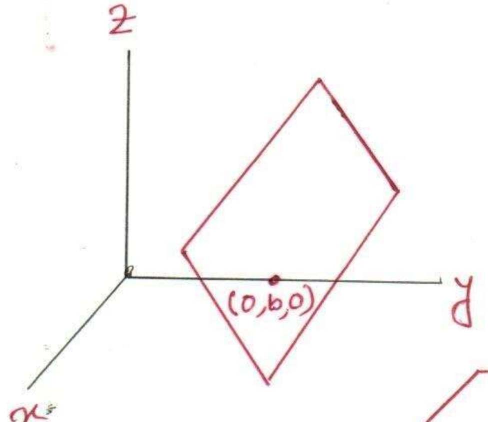


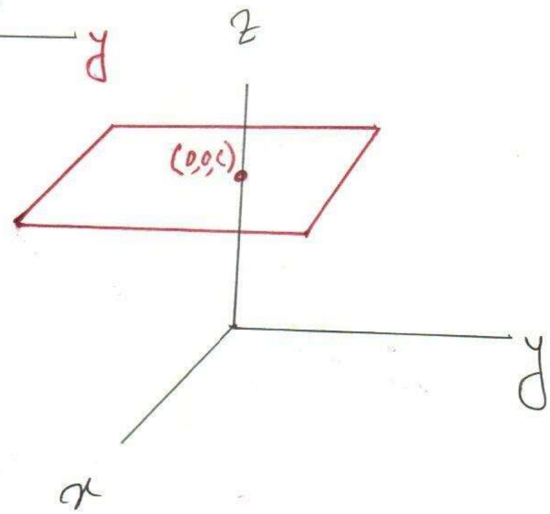
Equations of the planes parallel to coordinate planes

Any point on a plane parallel to YZ-plane at a distance a from it will have its x-coordinate as a. Therefore, the eqn $x = a$ represents a plane parallel to YZ-plane and at a distance a from it.

Similarly, the eqn of the plane parallel to ZX-plane at a distance b from it is $y = b$ and that of the plane parallel to xy-plane at a distance c from it is $z = c$.







Angle between two planes: Angle between two ~~poin~~planes is equal to the angle between the normals to them from any point. Thus the angle between the two planes

$$a_1x + b_1y + c_1z + d = 0 \quad -(1)$$
$$\text{and } a_2x + b_2y + c_2z + d = 0 \quad -(2)$$

is equal to the angle between the lines with direction ratios a, b, c and a_1, b_1, c_1. Hence if θ be the angle between the normals, then

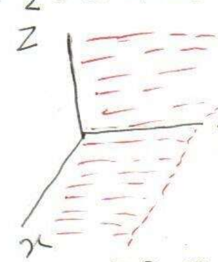
$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\ \sqrt{a_2^2+b_2^2+c_2^2}}$$

Ex:

Condition of perpendicularity of two planes.

If two planes $a_1x + b_1y + c_1z + d = 0$ and $a_2x + b_2y + c_2z + d = 0$ are perpendicular to each other, so their normals whose direction cosines are proportional to a_1, b_1, c_1 and a_2, b_2, c_2. Hence the condition of perpendicularity is $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Ex: Show that the planes $x = 0$ and $z = 0$ are perpendicular.



Soln: $x = 0 \Rightarrow 1\cdot x + 0\cdot y + 0\cdot z = 0$

$z = 0 \Rightarrow 0\cdot x + 0\cdot y + 1\cdot z = 0$

Now $1\cdot 0 + 0\cdot 0 + 0\cdot 1 = 0$

So the planes $x = 0$ and $z = 0$ are perpendicular.

Condition of parallelism of two planes

Two planes $a_1x + b_1y + c_1z + d = 0$ and $a_2x + b_2y + c_2z + d = 0$ are parallel iff the co-efficients of x, y, z in their eqns are proportional i.e. if $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$.

Some System of Plane

① Plane parallel to a given plane and passing through a given point.

The eqn of a plane parallel to a given plane $ax+by+cz+d=0$ is of the form

$$ax+by+cz+K=0 \quad \text{--- (i)}$$

Where K is an arbitrary constant. If it passes through a point (x_0, y_0, z_0), then (i) becomes —

$$ax_0+by_0+cz_0+K=0 \quad \text{--- (ii)}$$

Now (i) − (ii) ⇒

$$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$

Which is the required eqn of the plane.

Ex: Find the eqn of the plane through the point (2,1,1) and parallel to the plane $3x+2y+z-7=0$.

Soln:- The eqn of any plane parallel to the given plane $3x+2y+z-7=0$ is

$$3x+2y+z+K=0 \quad \text{--- (i)}$$

Where K is an arbitrary constant.

Since the plane passes through the point (2,1,1), we have $3\cdot2+2\cdot1+1+K=0 \Rightarrow K=-9$.

So (i) ⇒ $3x+2y+z-9=0$ which is the required eqn of the plane.

OR $3(x-2)+2(y-1)+1(z-1)=0$

$\Rightarrow 3x-6+2y-2+z-1=0$

$\Rightarrow 3x+2y+z-9=0$

(ii) Plane passing through two given points (x_1, y_1, z_1) and (x_2, y_2, z_2) and perpendicular to a given plane $ax + by + cz + d = 0$

Consider the required plane

$$a_1 x + b_1 y + c_1 z + d_1 = 0 \quad \text{--- (1)}$$

Since (1) passes through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) so $a_1 x_1 + b_1 y_1 + c_1 z_1 + d_1 = 0$ --- (2)

and $a_1 x_2 + b_1 y_2 + c_1 z_2 + d_2 = 0$ --- (3)

(2) − (1) and (3) − (1) give

$$a_1(x - x_1) + b_1(y - y_1) + c_1(z - z_1) = 0 \quad \text{--- (4)}$$

$$a_1(x - x_2) + b_1(y - y_2) + c_1(z - z_2) = 0 \quad \text{--- (5)}$$

Again if plane (1) is $\perp$ to the given plane then

$$a a_1 + b b_1 + c c_1 = 0 \quad \text{--- (6)}$$

Now eliminiating a_1, b_1, c_1 from eqns (4), (5) and (6) we get

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a & b & c \end{vmatrix} = 0$$

Which is the required eqn of plane.

Ex: Find the eqn of the plane through the points $(2, 2, 1)$ and $(9, 3, 6)$ and $\perp$ to the plane $2x + 6y + 6z - 9 = 0$.

Soln: The required eqn is

$$\begin{vmatrix} x - 2 & y - 2 & z - 1 \\ 2 - 9 & 2 - 3 & 1 - 6 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

$\Rightarrow 3x + 4y - 5z = 9$

Ex: Find the eqn of the plane which passes through the points $(1,0,-1)$ and $(2,1,3)$ and is perpendicular to the plane $2x+y+z=1$.

Ans: $3x-7y+z=2$

Cor. Planes perpendicular to coordinates planes

We want to find the planes which are $\perp$ to yz plane i.e, $x=0$, zx-plane i.e $y=0$ and xy plane i.e, $z=0$.

For yz plane i.e $x=0 \Rightarrow 1.x+0.y+0.z=0$ —— (i)

Let the eqn of plane $\perp$ to yz plane be

$$ax+by+cz+d=0 \quad \text{—— (ii)}$$

Since plane (i) and plane (ii) are $\perp$, so

$$1.a+0.b+0.c=0 \Rightarrow a=0.$$

Hence the eqn of the plane $\perp$ to yz plane is

$$by+cz+d=0$$

similarly the eqn of the planes $\perp$ to zx and xy planes are $ax+cz+d=0$ and $ax+by+d=0$ respectively.

Plane Trough the line intersection of given planes

Let $a_1x+b_1y+c_1z+d_1=0$ —— (1)

and $a_2x+b_2y+c_2z+d_2=0$ —— (2)

be two given planes. Then the eqn of plane passing through the line of intersection of the planes (1) and (2) is

$$a_1x+b_1y+c_2z+d_1+K(a_2x+b_2y+c_2z+d_2)=0 \quad \text{—— (3)}$$

Where $K(\neq 0)$ is an arbitrary constant.

The value of K is obtained if the plane (3) is $\parallel$ or $\perp$ to a given plane.

Ex Find the eqⁿ of the plane which is perpendicular to the plane $5x+3y+6z+8=0$ and which contains the line of intersection of the planes

$$x+2y+3z-4=0 \text{ and } 2x+y-z+5=0.$$

Solⁿ:- The eqⁿ of any plane through the line of intersection of the planes $x+2y+3z-4=0$ and $2x+y-z+5=0$ is

$$x+2y+3z-4+K(2x+y-z+5)=0$$

$$\Rightarrow (1+2K)x+(2+K)y+(3-K)z+(5K-4)=0 \quad \text{--- } (1)$$

where K is a parameter.

If the plane (1) is $\perp$ to $5x+3y+6z+8=0$, then

$$(1+2K)\cdot 5+(2+K)\cdot 3+(3-K)\cdot 6=0$$

$$\Rightarrow 7K+29=0 \Rightarrow K=-\frac{29}{7}.$$

Putting the value of K in (1), we have

$$\left(1-2\cdot\frac{29}{7}\right)x+\left(2-\frac{29}{7}\right)y+\left(3+\frac{29}{7}\right)z+\left(-5\cdot\frac{29}{7}-4\right)=0$$

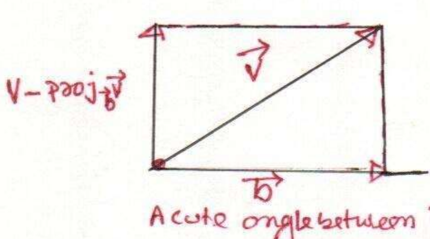
$$\Rightarrow \frac{-51}{7}x-\frac{15}{7}y+\frac{50}{7}z-\frac{173}{7}=0$$

$$\Rightarrow 51x+15y-50z+173=0.$$

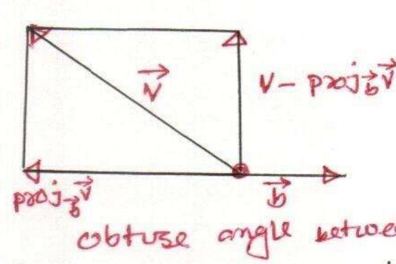
Perpendicular distance of the point (x_1, y_1, z_1) from the plane $ax+by+cz+d=0$

The orthogonal projection of $\vec{v}$ on an arbitrary nonzero vector $\vec{b}$ is

$$\boxed{\text{Proj}_{\vec{b}}\vec{v}=\frac{\vec{v}\cdot\vec{b}}{\|\vec{b}\|^2}\vec{b}}$$



Acute angle between $\vec{v}$ & $\vec{b}$

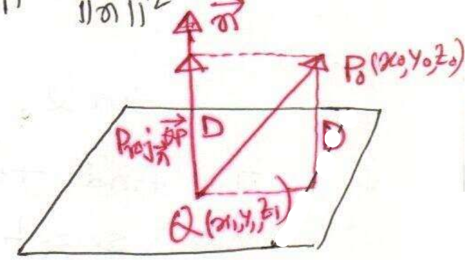


obtuse angle between $\vec{v}$ and $\vec{b}$

Geometrically, if $\vec{b}$ and $\vec{v}$ have a common initial point, then $\text{proj}_{\vec{b}}\vec{v}$ is the vector that is determined when a perpendicular is dropped from the terminal point of $\vec{v}$ to the line through $\vec{b}$.

Proof: Let $Q(x_1, y_1, z_1)$ be any point in the plane, and the position the normal $\vec{n} = \langle a, b, c \rangle$, so that its initial is at $Q(x_1, y_1, z_1)$. As shown in fig, the distance D is equal to the length of the orthogonal projection of $\overrightarrow{QP_0}$ on $\vec{n}$.

So
$$D = \| \text{Proj}_{\vec{n}} \overrightarrow{QP_0} \| = \left\| \frac{\overrightarrow{QP_0} \cdot \vec{n}}{\|\vec{n}\|^2} \vec{n} \right\| = \frac{|\overrightarrow{QP_0} \cdot \vec{n}|}{\|\vec{n}\|^2} \|\vec{n}\|$$
$$= \frac{|\overrightarrow{QP_0} \cdot \vec{n}|}{\|\vec{n}\|}$$



But $\overrightarrow{QP_0} = \langle x_0 - x_1, y_0 - y_1, z_0 - z_1 \rangle$

$\therefore \overrightarrow{QP_0} \cdot \vec{n} = a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)$ ax+by+cz+d=0

$\|\vec{n}\| = \sqrt{a^2 + b^2 + c^2}$

$\therefore$ Thus
$$D = \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}}$$
$$= \frac{|ax_0 + by_0 + cz_0 - (ax_1 + by_1 + cz_1)|}{\sqrt{a^2 + b^2 + c^2}}.$$

Since the point $Q(x_1, y_1, z_1)$ lies in the plane,

So $ax_1 + by_1 + cz_1 + d = 0 \Rightarrow ax_1 + by_1 + cz_1 = -d$

$$\therefore D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Ex: Find the distance D between the point $(1, -4, -3)$ and the plane $2x - 3y + 6z + 1 = 0$.

Soln: We know the distance from the point (x_0, y_0, z_0) to the plane $ax + by + cz + d = 0$ is
$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

So
$$D = \frac{|2 \cdot 1 + (-3) \cdot (-4) + 6 \cdot (-3) + 1|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{|-3|}{7} = \frac{3}{7}.$$

Perpendicular distance of the point $P(x_1, y_1, z_1)$ from the plane

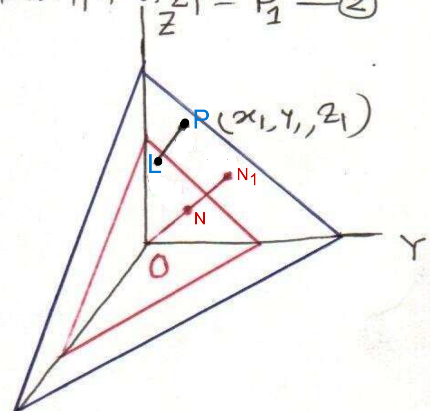
$lx+my+nz=p$, where p stands for the length of the perpendicular from the origin i.e. $ON=p$.

The given eqn of plane is $lx+my+nz=p$ —— (1)

Now we are interested to find the perpendicular distance of the plane from a point $P(x_1, y_1, z_1)$

The eqn of any plane parallel to (1) through P is

$$lx_1+my_1+nz_1=p_1 \text{ —— (2)}$$



Where $ON_1=p_1$ is the length of the perpendicular from the origin to plane (2) i.e the plane through $P(x_1, y_1, z_1)$ and parallel to the given plane (1)

Now draw (D=) PL perpendicular from the point P to the given plane. ON is the perpendicular from the origin to the given plane.

$$\therefore PL = D = ON_1 - ON = p_1 - p = lx_1+my_1+nz_1-p$$

Therefore, the length of the perpendicular from a given point $P(x_1, y_1, z_1)$ to $lx+my+nz=p$ is

$$D = lx_1+my_1+nz_1-p.$$

Angle between two planes
Bisectors of angles between two planes

Angle between two planes is equal to the angle between the normals to them from any point.

Thus the angle between the two planes

$$ax+by+cz+d=0 \text{ —— (1)}$$

and $a_1x+b_1y+c_1z+d_1=0$ —— (2)

is equal to the angle between the lines with direction ratios a, b, c and a_1, b_1, c_1. Hence if θ be the angle between the normals, then

$$\cos\theta = \frac{aa_1+bb_1+cc_1}{\sqrt{(a^2+b^2+c^2)}\,\sqrt{(a_1^2+b_1^2+c_1^2)}}$$

Ex: Find the acute angle between the planes -

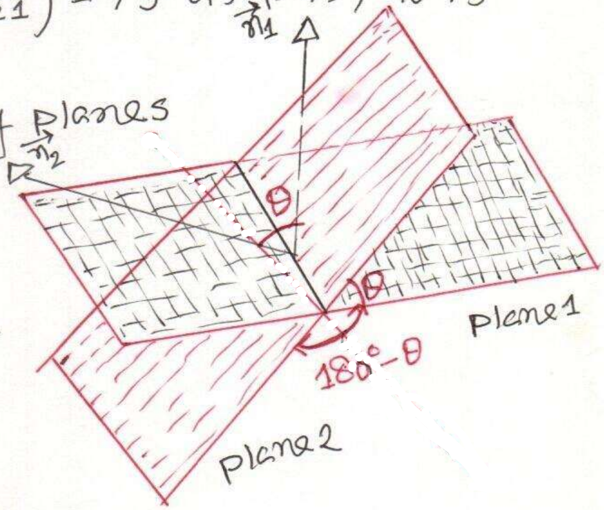
$$2x-4y+4z=6 \quad \text{---}(1)$$
$$\text{and } 6x+2y-3z=4 \quad \text{---}(2)$$

Sol^n:- The given eq^n yields the normals $\vec{n}_1 = \langle 2,-4,4\rangle$ and $\vec{n}_2 = \langle 6,2,-3\rangle$. Thus

$$\cos\theta = \frac{|\vec{n}_1\cdot\vec{n}_2|}{\|\vec{n}_1\|\,\|\vec{n}_2\|} = \frac{|2\cdot 6+(-4)\cdot 2+4\cdot(-3)|}{\sqrt{(2^2+(-4)^2+4^2)}\sqrt{(6^2+2^2+(-3)^2)}} =$$

$$= \frac{|-8|}{\sqrt{36}\sqrt{49}} = \frac{8}{6\cdot 7} = \frac{4}{21}.$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{4}{21}\right) = 79.01942457 \approx 79^0$$



Angle between two intersecting planes

Two distinct intersecting planes determine two positive angles of intersection - an (acute) angle θ that satisfies the condition $0\le\theta\le\pi/2$ and the suppliment of that angle.

If $\vec{n}_1$ and $\vec{n}_2$ are normals to the planes, then depending on the directions of $\vec{n}_1$ and $\vec{n}_2$, the angle θ is either the angle between $\vec{n}_1$ and $\vec{n}_2$ or the angle between $\vec{n}_1$ and $-\vec{n}_2$. In both cases, the following formula for the acute angle θ between the planes:-

$$\cos\theta = \frac{|\vec{n}_1\cdot\vec{n}_2|}{\|\vec{n}_1\|\,\|\vec{n}_2\|}$$

[E] Bisectors of angles between two planes

Let the given two planes are

$$a_1x + b_1y + c_1z + d_1 = 0 \quad \text{--- (1)}$$
$$a_2x + b_2y + c_2z + d_2 = 0 \quad \text{--- (2)}$$

Let (x, y, z) be a point on any one of the bisecting planes. Then the perpendicular distance from this point to the two planes must be equal in magnitude

$$\therefore \frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Which give the eq^{n}s of the two bisecting planes.

Note1: If $a_1a_2 + b_1b_2 + c_1c_2 =$ negative, then the origin lie in the acute angle between the given planes.

Note2: If $a_1a_2 + b_1b_2 + c_1c_2 =$ positive, then the origin " " " obtuse angle " " " ".

Ex: Show that the origin lies in the acute angle between the planes $x+2y+2z-9=0$ and $4x-3y+12z+13=0$. Find the planes bisecting the angles between them and point out which bisects the acute angle.

Sol^n:- The given planes are $-x-2y-2z+9=0$ --- (1)
and $4x-3y+12z+13=0$ --- (2)

Here $a_1a_2 + b_1b_2 + c_1c_2 = (-1)\cdot 4 + (-2)(-3) + (-2)\cdot 12$
$= -4 + 6 - 24 = -22 < 0$

Hence the origin lies in the acute angle between the given planes.

Now the eq^n of the planes bisecting the angles between the given planes (1) and (2) are

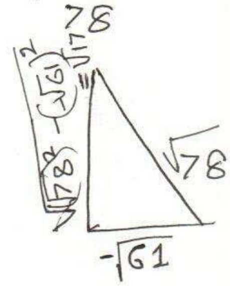
$$\frac{-x-2y-2z+9}{\sqrt{(-1)^2+(-2)^2+(-2)^2}} = \pm \frac{4x-3y+12z+13}{\sqrt{4^2+(-3)^2+(12)^2}}$$

$\Rightarrow 25x + 17y + 62z - 78 = 0$ or $x + 35y - 10z - 156 = 0$

Let θ be the angle between the planes

$$-x-2y-2z+9=0 \text{ and } 25x+17y+62z-78=0, \text{ then}$$

$$\cos\theta = \frac{(-1)\cdot 25+(-2)\cdot 17+(-2)\cdot 62}{\sqrt{(-1)^2+(-2)^2+(-2)^2}\sqrt{25^2+17^2+62^2}} = -\sqrt{\frac{61}{78}}$$



$$\therefore \tan\theta = -\frac{\sqrt{17}}{\sqrt{61}} < 1$$

that is, $\tan\theta < 1$

$$\Rightarrow \theta < \pi/4$$

Thus the plane $25x+17y+62z-78=0$ bisects the acute angle between the two given planes.

Ex: Show that the plane $14x-8y+13=0$ bisects the obtuse angle between the planes $3x+4y-5z+1=0$ and $5x+12y-13z=0$

soln:- The given eqn of the planes are

$$3x+4y-5z+1=0 \quad \text{——} (1)$$

and

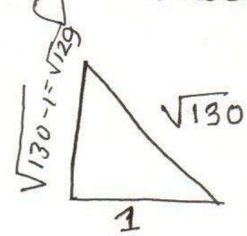
$$5x+12y-13z=0 \quad \text{——} (2)$$

Here $a_1a_2+b_1b_2+c_1c_2 = 3\cdot 5+4\cdot 12+(-5)(-13) = 180 > 0$

So the origin lies in the obtuse angle between the planes.

The eqns of the planes bisecting the angles between the given planes (1) and (2) are

$$\frac{3x+4y-5z+1}{\sqrt{3^2+4^2+(-5)^2}} = \pm\frac{5x+12y-13z}{\sqrt{5^2+12^2+(-13)^2}}$$



$$\Rightarrow 14x-8y+13=0 \quad \text{—}(3) \quad \text{or} \quad 64x+112y-130z+13=0 \quad \text{—}(4)$$

Let θ be the angle between (1) and (3), then

$$\cos\theta = \frac{3\cdot 14+4\cdot(-8)+(-5)\cdot 0}{\sqrt{3^2+4^2+(-5)^2}\cdot\sqrt{14^2+(-8)^2}} = \frac{10}{\sqrt{50}\sqrt{260}} = \frac{1}{\sqrt{130}}$$

$$\therefore \tan\theta = \sqrt{129} > 1 \Rightarrow \theta > \pi/4$$

Hence $14x-8y+13=0$ bisects the obtuse angle between the two given planes.

4. Show that the general homogeneous eqn of the 2nd degree $ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$ will represent a pair of planes if

$$abc+2fgh-af^2-bg^2-ch^2=0$$

Proof: The general homogeneous eqn of 2nd degree is

$$cz^2+ax^2+by^2+2fyz+2gzx+2hxy=0 \quad (1)$$

$$\Rightarrow ax^2+2(hy+gz)x+(by^2+cz^2+2fyz)=0 \quad (2)$$

Which is a quadratic eqn in x.

Solving (2) for x, we have

$$x=\frac{-2(hy+gz)\pm\sqrt{4(hy+gz)^2-4a(by^2+cz^2+2fyz)}}{2a}$$

$$=\frac{-(hy+gz)\pm\sqrt{(hy+gz)^2-a(by^2+cz^2+2fyz)}}{a}$$

Eqn (2) will represent a pair of plane if it can be solved into two linear factors, that is if the expression under ~~radi~~ square root (radical sign) is a perfect square i.e. if

$(hy+gz)^2-a(by^2+cz^2+2fyz)$ is a perfect square.

i.e. $(h^2-ab)y^2+2yz(gh-af)+(g^2-ac)z^2$ " "

i.e. $\{2z(gh-af)\}^2-4(h^2-ab)(g^2-ac)z^2=0$

$$\Rightarrow (gh-af)^2-(h^2-ab)(g^2-ac)=0$$

$$\Rightarrow \cancel{g^2h^2}-2agfh+a^2f^2-\cancel{h^2g^2}+h^2ac+abg^2-a^2bc=0$$

$$\Rightarrow abc+2fgh-af^2-bg^2-ch^2=0$$

Area of a triangle whose vertices are $(x_1, y_1, z_1), (x_2 y_2 z_2)$ and (x_3, y_3, z_3) is

$$A = \frac{1}{2}\sqrt{\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 + \begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}^2 + \begin{vmatrix} z_1 & x_1 & 1 \\ z_2 & x_2 & 1 \\ z_3 & x_3 & 1 \end{vmatrix}^2}$$

Volume of a tetrahedron (চতুস্তলক) whose four vertices are $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$ and $D(x_4, y_4, z_4)$ is

$$V = \frac{1}{6}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$$

Volume of a tetrahedron (চতুস্তলক) when its four faces are given by $a_r x + b_r y + c_r z + d_r = 0$; $r = 1, 2, 3, 4$ is

$$V = \frac{\Delta^3}{6 D_1 D_2 D_3 D_4} \quad \text{Where} \quad \Delta = \begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix}$$

and D_1, D_2, D_3, D_4 are the cofactors of d_1, d_2, d_3, d_4 respectively in the determinant Δ.

Ex: A variable plane is at a constant distance p from the origin and meets the axis in A, B, C. Through A, B, C planes are drawn parallel to the co-ordinate planes. Show that the locus of their point of intersection is $x^{-2} + y^{-2} + z^{-2} = p^{-2}$.

Soln: Let the eqn of the variable plane be

$$lx + my + nz = p$$

Here l, m, n vary but p is constant as the plane is always at constant distance from the origin. l, m, n are the actual direction cosines of the plane.

It cuts the axis of x at $A\left(\frac{p}{l}, 0, 0\right)$, axis of y at $B\left(0, \frac{p}{m}, 0\right)$ and axis of z at $C\left(0, 0, \frac{p}{n}\right)$.

Now the plane through $A\left(\frac{p}{l}, 0, 0\right)$, $B\left(0, \frac{p}{m}, 0\right)$, $C\left(0, 0, \frac{p}{n}\right)$ are parallel to the co-ordinate planes, Then the co-ordinate of point of intersection $\left(\frac{p}{l}, \frac{p}{m}, \frac{p}{n}\right)$.

$$\therefore \; x = \frac{p}{l}, \; y = \frac{p}{m}, \; z = \frac{p}{n}.$$

Therefore $\frac{p^2}{x^2} + \frac{p^2}{y^2} + \frac{p^2}{z^2} = l^2 + m^2 + n^2 = 1$

$$\Rightarrow x^{-2} + y^{-2} + z^{-2} = p^{-2}.$$

EX: A variable plane is at a constant distance p from the origin and meets the axes in A, B, C. Show that the locus of the centroid of the tetrahedorn OABC is $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$.

Theorem: If a, b, c, and d are constant, and a, b, and c are not all zero, then the graph of the eq^n

$$ax+by+cz+d=0 \quad \text{---} \quad (1)$$

is a plane that has the vector $\vec{n} = \langle a,b,c \rangle$ as a normal.

Proof: Since a, b and c are not all zero, there is at least one point (x_0, y_0, z_0) whose coordinates satisfy eq^n (1). For example, if $a \neq 0$, then such a point $\left(-\frac{d}{a}, 0, 0\right)$, and similarly if $b \neq 0$, then $\left(0, -\frac{d}{b}, 0\right)$, and if $c \neq 0$, then $\left(0, 0, -\frac{d}{c}\right)$ are satisfy the eq^n (1). Thus, Let (x_0, y_0, z_0) be any point whose co-ordinates satisfy (1) that

$$ax_0+by_0+cz_0+d=0 \quad \text{---} \quad (2)$$

$$(1) - (2) \Rightarrow a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$

Which is the point-normal form of a plane with normal $\vec{n} = \langle a, b, c \rangle$ ■

Eq^n (1) is called the general form of the eq^n of a plane

Ex: Determine whether the planes

$3x-4y+5z=0$ and $-6x+8y-10z-4=0$ are parallel.

Sol^n: - It is clear geometrically that two planes are parallel iff their normals are parallel vectors.

A normal to the first plane is $\vec{n}_1 = \langle 3,-4,5 \rangle$

and a " " " 2nd " " $\vec{n}_2 = \langle -6, 8, -10 \rangle$

$$= -2\langle 3,-4,5 \rangle$$

$$= -2\,\vec{n}_1$$

Since $\vec{n}_2$ is a scalar multiple of $\vec{n}_1$, the normals are parallel, and hence so are the planes

OR Since $\frac{3}{-6} = \frac{-4}{8} = \frac{5}{-10}$, so the two planes are $\parallel$.